

CANVAS EXPORT

System flow assessment

Product completeness of DECSI Loan Hub against the 12-stage diagram in presentation/system_flow.png. Code and automated-test review — not a live production UAT at DECSI.

90.5% flow completeness

Mean of 12 stage scores (0–100). 1,086 / 1,200 completeness points.

9 / 12 fully implemented

Application through appraisal, authorization, workflow, and analytics.

1 config · 2 partial

Disbursement mock until CBS. Legal admin and dedicated fraud still partial.

250 / 250 tests

Clean gate after 20 Aug 2026 fixes. docker compose exec web python manage.py test

SOURCE CANVAS

system-flow-assessment.canvas.tsx

DATE

20 August 2026

DIAGRAM

presentation/system_flow.png

CLASSIFICATION

Seqela internal — do not send as-is

1. Overall

The credit factory is built. Weighted completeness is 90.5% across the 12 flowchart boxes. Nine stages are fully implemented in product code. Disbursement to core banking is coded but mock-by-default until DECSI_BASE_URL is set. Two boxes are only partial: standalone legal administration, and a dedicated fraud-detection product.

90.5%

Flow completeness

9/12

Fully implemented

2

Partial stages

250/250

Automated tests passed

Coverage of the 12-box flow



Green = fully implemented scores (854) · Blue = CBS pending (82) · Gold = partial (150) · 1,086 / 1,200 points · 90.5% of flowchart scope.

Stage verdict mix

9

Fully implemented

1

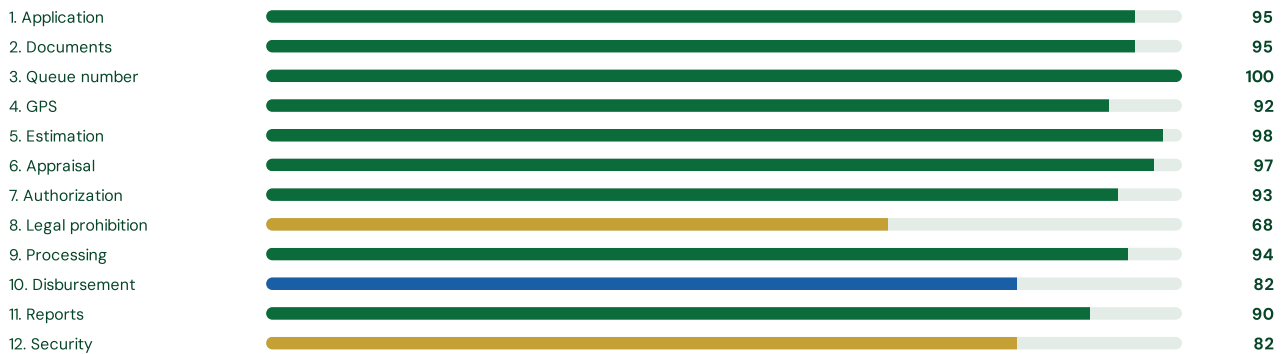
Live CBS pending

2

Partial

Count of flowchart boxes · n = 12 · Source: code review 20 Aug 2026.

Completeness by stage (%)



Score 0–100 per flowchart box · mean 90.5% (dashed target on canvas). Gold = partial · blue = config.

2. Stage-by-stage result

Scores are reviewer estimates of flowchart intent vs shipped code, not production KPIs.

#	FLOWCHART BOX	VERDICT	SCORE	WHAT THE PRODUCT DOES	EVIDENCE	GAP
1	Application Applying for loan; getting information	FULLY IMPLEMENTED	95	Staff create at /hub/create_loan_request/ plus Digital Apply at /apply/new/ (customer lookup, loan types, online intake)	LoanRequest, OnlineApplication, LoanCategory; submit_online_application	CBS customer lookup falls back to mock when DECSI_BASE_URL is unset
2	Documents Necessary documents; get approval	FULLY IMPLEMENTED	95	Per-loan-type checklist, Eng+Amh OCR, integrity/identity checks, officer verify/reject	LoanRequestDocument.auth_status; document_auth.py; /hub/loan_request/<id>/documents/	LLM and external ID checks are optional hooks, not always on
3	Loan Queue No Generation	FULLY IMPLEMENTED	100	HK-##### issued on staff, online, agent, and CSV create	generate_incremental_loan_request_id(); LatestLoanRequestID	No separate ticket-printer app — the queue number is loan_request_id
4	GPS Collateral integration; data capture	FULLY IMPLEMENTED	92	Field visits for building/land/other: site GPS, photo EXIF, maps, offline tablet sync, evidence pack	collateral/views field_visit; site_gps_lat/lon; /collateral/loan/<id>/...	Gebeta tiles need an API key; OSM fallback works without it
5	Collateral Estimation Estimation detail; estimated amount	FULLY IMPLEMENTED	98	BOQ catalog, unit prices, building/land/movable valuation, engineering QA, grand total into appraisal Sheet 5	BuildingValuation, LandValuation, compute_collateral_totals; /collateral/engineering-qa/	None material
6	Loan Appraisal Financial, business, approval	FULLY IMPLEMENTED	97	7-step MSME and corporate wizard: character, cashflow/ratios, E&S, collateral, recommendation, optional schedule	APPRAISAL_STEPS; /hub/loan_request/<id>/appraisal/step/<n>/	Sheet 7 (repayment schedule) is optional by design

#	FLOWCHART BOX	VERDICT	SCORE	WHAT THE PRODUCT DOES	EVIDENCE	GAP
7	Authorization Branch / zonal / HO operation / HO mgt	FULLY IMPLEMENTED	93	Configurable committees: branch → district → head_office → management, amount bands, votes, delegations	ApprovalCommitteeLevel keys; loans/committee.py; /hub/manage_approval_committees/	Flowchart “Zonal” is District; “HO Operation” is head_office (Cooperative intake is a separate gate)
8	Prohibition of collateral in legal administration	PARTIALLY IMPLEMENTED	68	Post-approval legal papers that block disbursement: restriction, POA, title search, mortgage, notary	loans/collateral_legal.py; /hub/loan_request/<id>/post_approval/	No Legal role, legal queue, or legal_cleared stage — enforced as a finance gate, not its own module
9	Processing Workflow Status tracking; integration & verification	FULLY IMPLEMENTED	94	Role queues, committee status, document verify, process policy, risk desk, notifications	committee_status / disbursement_status; process_policy.py; /hub/view_loan_requests/	Workflow is Django state machines, not a BPMN engine
10	Disbursement Disbursing to core banking	CODE COMPLETE, LIVE CBS PENDING	82	Post-approval → Finance → confirm → book. Live Temenos client exists; mock ledger is the default	loans/disbursement.py; cbs_client.py; DECSI_CBS_BOOK_ON_DISBURSE	Live booking needs DECSI_BASE_URL; DECSI_CBS_USE MOCK_LEDGER defaults to True
11	Report & Analytics	FULLY IMPLEMENTED	90	Classic reports plus Credit Intelligence (pipeline, collateral, decision, assistant) and book-ops desks	/hub/credit-intelligence/; reporting.py; views_book.py	NPL / outstanding KPIs stay adapter-backed and weak until live CBS
12	Security and Compliance RBAC; authorization; fraud detection	PARTIALLY IMPLEMENTED	82	20+ roles, MFA, lockout, audit log, committee bands, password min-length 10, nosniff/DENY headers, document “suspicious” verdicts, watchlist, risk desk	CustomUser.ROLE_CHOICES; EnforceMfaMiddleware; SecurityAuditLog; /hub/security/audit/	No dedicated fraud/AML case engine or transaction-fraud model — document auth + watchlist only

3. Flowchart labels vs product names

Authorization titles in the diagram do not match the committee keys in code. Functionally the four-level chain is there.

SYSTEM_FLOW.PNG	PRODUCT IMPLEMENTATION	MATCH
Branch (S/branch) Level	ApprovalCommitteeLevel key = branch (sub-branch via Branch)	Direct
Zonal Level	key = district (District Manager / district committee)	Renamed
HO Operation Level	key = head_office committee; Cooperative intake is a separate queue	Split
HO Mgt Level	key = management (CEO / VP / Board)	Direct
Prohibition of collateral in legal administration	Post-approval legal documents + disbursement blockers	Relocated
Disbursing to Core Banking system	cbs_client + Finance gate; mock ledger unless DECSI_BASE_URL	Config
Fraud Detection	Document OCR suspicious verdict + watchlist + risk desk	Partial

4. Gaps that keep the flow from 100%

8. Legal administration is a finance gate, not a module · 68%

The diagram places prohibition of collateral after authorization as its own step. The product stores restriction, power of attorney, title search, mortgage registration, and notary papers on the post-approval screen and blocks mark-disbursed until they are verified. There is no lawyer role, legal queue, or legal_cleared status. Closing the gap means a Legal Administration queue between committee_approved and Finance.

12. Fraud detection is document-level, not a fraud product · 82%

Role-based access, committee authorization bands, TOTP MFA, session logout, password minimum length 10, nosniff / X-Frame-Options DENY, and SecurityAuditLog are implemented. What the flowchart calls fraud detection is OCR integrity, identity match, a suspicious document verdict, a loan watchlist, and the risk desk. There is no AML case system, device/velocity engine, or transaction-fraud model.

10. Core banking disbursement is mock until DECSI_BASE_URL · 82%

The path post-approval → Finance approval → officer confirm → CBS book is fully coded (cbs_client.py, portfolio_ledger.py, tests). Default settings use DECSI_CBS_USE MOCK_LEDGER=True. Live Temenos booking requires DECSI_BASE_URL. Customer lookup at application time has the same mock fallback.

Adjacent: Seqela Market staff admin is unwired · portal only

The external market portal at /market-portal/ is live. Staff market admin views exist under partners/ but are not mounted in the root URL conf (tests expect /partners/ to 404). This is outside the 12 boxes but is part of the surrounding product.

5. Automated test run

Clean gate: 250 / 250 passed. Re-run in the web container on 20 Aug 2026 after the seven failures/errors were fixed. python manage.py test discovers the loans.tests package (empty loans/tests.py stub removed).

250

Tests run

250

Passed

0

Failed

35.2s

Duration

PREVIOUS ISSUE	RESOLUTION
Offline SW test still expected cache v8; help-text assertion stale	Test matches current collateral-field-vN cache and sync/retry bar
Appraisal lock POST hit /loan_request/... and got HTTP 301	Legacy hub redirect is 302; test posts via reverse() to /hub/...
Credit LO HO create used customer number C-1 (too short)	Test uses mock customer 2000050041; create path returns 302
Security tests: missing _env_bool, nosniff header, password min length	settings.py now has the helper, always-on headers, PASSWORD_MIN_LENGTH=10

Pass rate 100% (250/250) · docker compose exec web python manage.py test

6. Closing the flowchart

1. Add a Legal Administration queue and role so stage 8 sits between committee approval and Finance, matching the diagram.
2. Point DECSI_BASE_URL at core banking and turn off mock ledger in the target environment so stage 10 is a live disbursement.
3. If the diagram's "Fraud Detection" is a contractual requirement, specify the product (document-only vs AML/transaction) — current coverage is origination fraud controls only.
4. Automated tests are a clean gate (250/250). Remaining work is process design (legal desk, fraud product) and CBS go-live, not test repair.

Not scored here: DECSI production currently runs queue and admin only; this assessment is of the enhanced credit-factory codebase against system_flow.png.



Source diagram: presentation/system_flow.png

Export of Cursor canvas system-flow-assessment.canvas.tsx · 20 August 2026 · Seqela internal.
Client-facing pack (no scores / file paths): DECSI_System_Flow_Assessment.pdf